



PERSONAL INFO

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GITHUB

github.com/grzegorzczy

LINKEDIN

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LOCATION

Wrocław, Poland

SKILLS

PLC & Automation: Siemens (TIA Portal), JUMO variTRON (CODESYS), HMI/SCADA, ST, LAD, FBD

Industrial IT & IoT: OPC UA, MQTT, EtherNet/IP, Modbus RTU/TCP, PROFINET, Node-RED

Software Engineering: Python, SQL, n8n, Git/GitHub, Docker, CI/CD

ML & Computer Vision: PyTorch, scikit-learn, OpenCV, MediaPipe, time-series forecasting, anomaly detection

Other: MS Office, Driving License (Cat. B), SEP Electrical Qualification (up to 1 kV)

LANGUAGES

Polish	Native
English	B2
German	A2

COURSES & CERTIFICATION

Python in Industrial Automation

Programming Siemens S7-1200 / S7-1500 PLCs using Structured Control Language (SCL)

Machine Learning Course

Grzegorz Czyrniański

APPLICATION DEVELOPER

Application Developer at JUMO with a Master's degree in IT Automation Systems and hands-on experience executing industrial projects end-to-end, from system design to commissioning. I bridge traditional Operational Technology (PLC programming, industrial networks) and modern IT infrastructure. My goal is to develop and deploy AI solutions — particularly in computer vision, object detection, and anomaly detection — for complex Industry 4.0 challenges.

WORK HISTORY

Application Developer

2025-08 – present

JUMO, Wrocław

- Collaborating with international engineering teams (incl. direct cooperation with the Austrian branch) to design and commission end-to-end automation solutions.
- Developing and optimizing advanced PLC software for complex industrial processes, ensuring high system reliability.
- Establishing robust industrial networks and cross-platform communication architectures utilizing OPC UA, MQTT, and PROFINET.
- Leading the modernization of control systems by migrating legacy hardware to modern PLC platforms, enhancing capabilities and IoT readiness.
- Designing intuitive HMI/SCADA visualization layers for real-time process monitoring, alarming, and data acquisition.

Inside Sales

2024-06 – 2025-08

JUMO, Wrocław

- Office-based role in the sales of process automation components, in close cooperation with external sales representatives.
- Selecting equipment suited to customer applications and preparing sales offers, including systems and solutions.
- Handling post-offer communication and participating in price negotiations.
- Close collaboration with the production department and subcontractors (technical arrangements, negotiations).

Internship Student

2023 – 2024-05

Electro – Electrical Installation and Measurement, Olawa

- Interpretation of electrical diagrams and technical documentation.
- Installation of electrical systems and wiring.

PROJECTS

Gesture-Controlled Package Sorting System (Computer Vision + PLC)

GitHub · Video demo

- Developed a real-time hand-gesture recognition pipeline (Python, OpenCV, MediaPipe) that counts extended fingers to select a routing path, controlling diverter gates that sort packages onto different conveyor belts in a Factory I/O simulation.
- Built the IT/OT integration layer using a Prosys OPC UA server and Node-RED to bridge the vision system and a Siemens PLC (TIA Portal) over OPC UA and the S7 protocol.

Multi-Agent AI Automation System (n8n)

- Built an autonomous multi-agent workflow in n8n using LLMs (Google Gemini, Perplexity) with multi-modal pipelines for image/audio analysis and web research.
- Integrated Telegram and Google Workspace APIs to deploy a functional product for user interaction and automated task scheduling.

Pasteurisation Process Monitoring and Control System

- Integrated wireless temperature sensors with a PLC for real-time data acquisition and embedded analytics (e.g., calculating Pasteurization Units – PU).
- Designed a SCADA/HMI layer for process visualization, alarming, and data archiving, structured for future cloud integration.

EDUCATION

M.Sc., IT Automation Systems

2024-03 – 2025-07

Wrocław University of Science and Technology

Specialization: Intelligent Industry Systems 4.0

B.Eng., Industrial Automation

2021-10 – 2024-02

Wrocław University of Science and Technology

Specialization: Automation of Machines, Vehicles and Devices